

Mathematics — Diagnostic 1

Elchin Zeynalli · Mathematics-10 · Units E01–E02 + last year's skills

Name: _____ Date: _____ Start time: _____ End time: _____

Show your working next to every question. It's fine to get things wrong — this test is to find out what to practise. No calculator. Around 35 minutes; if you don't know a question, write a ? and move on.

A. Numbers you learned last year

1. Write in digits: **four hundred six thousand, twenty-nine**.

2. Round 738,462 to the nearest ten thousand.

3. Work out $60,304 - 27,856$. Show your working.

4. Write the answers as fast as you can: $7 \times 8 = \underline{\quad}$ $6 \times 9 = \underline{\quad}$ $12 \times 7 = \underline{\quad}$ $8 \times 4 = \underline{\quad}$ $9 \times 9 = \underline{\quad}$
 $11 \times 12 = \underline{\quad}$

5. Work out $3,246 \times 7$.

6. Work out 46×23 .

7. Work out $2,957 \div 6$. Write the remainder.

8. List all the factors of 36. Then circle the ones that are prime.

9. Which is larger: $\frac{3}{4}$ or $\frac{5}{8}$? Explain how you know.

10. Work out $\frac{5}{6} - \frac{2}{6}$.

11. Write 0.47 as a fraction. Then write which is bigger: 0.47 or 0.5.

12. A box holds 24 pencils. A school buys 15 boxes and gives 9 pencils to each of 32 students.
How many pencils are left?

B. Place value and powers of ten

13. In the number 5,550, the 5 in the hundreds place is worth ____ and the 5 in the thousands place is worth _____. The thousands 5 is _____ times the hundreds 5.
14. Write 2,046,300,015 in words.
15. Write 830,207,000 in expanded form.
16. Put in order from smallest to largest: 4,090,000 4,900,000 4,009,000 4,099,000
17. Write $10 \times 10 \times 10 \times 10$ using an exponent, then write its value.
18. Work out: $37 \times 100 = \underline{\hspace{1cm}}$ $6 \times 10^3 = \underline{\hspace{1cm}}$ $4,500 \div 10^2 = \underline{\hspace{1cm}}$
19. Use a pattern to work out 30×400 . Explain the pattern.

C. Multiplication, division and properties

20. Fill in the blank using a property: $(7 \times 5) \times 2 = 7 \times (\underline{\hspace{1cm}} \times 2)$. Which property is this?
21. Use the distributive property to work out 6×48 by splitting 48.
22. Work out $4,825 \times 36$.
23. Kim says $5,208 \div 8 = 651$. Check her answer using multiplication. Is she right?

D. Expressions and order of operations

24. Write as a numerical expression (do not solve): **the sum of 12 and 8, multiplied by 3.**

25. Write in words what this expression means: $40 - (6 \times 5)$.

26. Work out $8 + 4 \times 3 - 2$.

27. Work out $36 \div (2 + 4) \times 5$.

28. Work out $[(18 - 6) \div 4] + 2 \times 7$.

29. Rasim worked out $5 + 3 \times 2$ and got 16. Explain in a sentence what he did wrong and what the correct answer is.